

Tusher Bhomik

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[Portfolio](#) | [LinkedIn](#) | [GitHub](#) | [LeetCode](#)

EDUCATION

• Bangladesh University of Engineering and Technology

Bachelor of Science in CSE

2022 – June 2026, CGPA: 3.3164/4.00

Dhaka, Bangladesh

RESEARCH

• Age-Aware Neurological Gait Pattern Discovery from Biomechanical Signals: Multi-Phase Stroke Severity Analysis

Undergraduate Thesis (defended) | Supervisor: Mahmuda Naznin, Dept. of CSE, BUET

2025
Dhaka, Bangladesh

- Built an **age-normative, leakage-controlled** framework for objective post-stroke gait severity on 138 healthy and 50 stroke adults, combining **aging-manifold residual z-scores**, ANCOVA speed adjustment, and an EMG-derived **Synergy Complexity Index** (Spearman $\rho = 0.532$ with clinical walking ability).
- Introduced a **leave-one-age-band-out** falsification test and a strict nested 10×5 -fold protocol that retracted an apparent SMOTE-vs-manifold augmentation effect as a **data-leakage** artefact.

• Reinforcement Learning for Reliable and Security-Aware LLM Agents (Ongoing)

2026 – Present

Supervisor: Md. Rizwan Parvez, QCRI Research Institute

Doha, Qatar

- Investigating reinforcement learning-based optimization to enhance the reliability and accuracy of function calling behavior in large language models for complex tool-use scenarios.
- Exploring reinforcement learning techniques for developing security-aware agentic systems capable of secure reasoning, vulnerability analysis, and autonomous decision making in software repositories.

PROJECTS

• HealthSync

Spring Boot, React, PostgreSQL, Firebase

2025

[GitHub](#)

- Developed a comprehensive healthcare platform connecting patients and doctors with appointment scheduling, digital prescription management, and medical history tracking.

• Arxiv Research Paper RAG System

Python, SQL, SQLite, FastAPI, ChromaDB, Data Analysis

2026

[GitHub](#)

- Built an end-to-end **data pipeline** for ingesting, cleaning, and analyzing arXiv research paper data using Python and SQL.
- Developed a **RAG-based API system** using FastAPI and ChromaDB for semantic retrieval and question answering.

• Enhanced xv6-riscv Kernel

C, Assembly, Makefile

2024

[GitHub](#)

- Implemented an **MLFQ scheduler** with aging and added **kernel-level threading** support.
- Added custom system calls (trace, sysinfo) and a Bash autograder for testing.

• Mini C Compiler

CSE-310 Compiler Design, C, 8086 Assembly

2024

[GitHub](#)

- Built a **mini C compiler** from scratch in four phases: Symbol Table, Lexical Analysis, Semantic Analysis, and Intermediate Code Generation.
- Generated **8086 assembly** as intermediate output to validate end-to-end compilation workflow.

ACHIEVEMENTS

• 5th Place — Tech Career Hunt 2.0, organized by Shohoj Coding.

May 2026

• Internship Selection at Save the Children — AI/ML Track Finalist in their Hackathon.

SKILLS

- **Research Interests:** Cyber-Physical Systems, Systems Security, Software Security, Trustworthy AI, AI Security Operating Systems, Reinforcement Learning, LLMs
- **Programming:** Python, C++, C, Java, JavaScript | **Databases:** MySQL, PostgreSQL, MongoDB
- **ML/AI:** PyTorch, TensorFlow, Keras, Scikit-learn, NumPy, Pandas, OpenCV, LLMs, vLLM
- **Tools:** Git, Docker, Linux | **Frontend:** React, Next.js, Material-UI

RELEVANT COURSEWORK

- **Relevant Coursework** Data Structure, Algorithm, Computer Networks, Database Management System, Compiler Design, Operating System, Software Engineering, Artificial Intelligence, Machine Learning, Computer Architecture, Theory of Computation, Computer Graphics

ADDITIONAL INFORMATION

Languages: English, Bengali. **Interests:** Photography, Traveling, Badminton, Cycling.